# Predictions of goal-scoring players: a comparative approach

Markov Chain Mavericks

The code may be found here

## 1 Introduction

Football Manager, a famous videogame in which the player pretends to be an actual football manager, often lets you answer in interviews that goals are the most important commodities for a team. Therefore, understanding the drivers behind footballer's scoring potential opportunities could lead to more informed choices by coaches who are willing to integrate their intuition with a data-driven approach.

The development of football analytics tools is currently a challenge that is not easy to tackle, as it is difficult to find objective metrics to the contribution of a player to the attacking or defensive phases of a match. Nevertheless, it is possible to identify some actions that, according to common sense, are able to outline a player's performance. In particular, Bransen et al. (2019) highlights that the quality of decisions and of the executions appear to have a geat influence on the outcome of a match.

Furthermore, Bunker and Thabtah (2019) shows that the most promising methodology involves the use of machine learning and neural networks, due to their high predictive power on historical datasets, despite the difficulties given by the small number of goals that are scored during matches.

The main purpose of this paper is to understand how well and which

player's on-field behaviour during a match can predict the probability that they will score a goal later in the game.

The outline of this paper is as follows. First, the data used in the analyses is presented. Section 3 will explain the methodologies applied to the data. Fianlly, the results will be described and discussed.

## 2 Data

The dataset in analysis contains 3486 rows, which represent players' activity snapshots at match quarter-hour checkpoints. Each row captures the player's activity in the 15 minutes before the checkpoint and cumulatively throughout the match, in addition to covariates such as the position and the fact whether a player was subbed in. It is important to notice that the dataset is largely imbalanced with an approximate ratio of 16 occurrences of the majority class with respect to one of the minority class, due to the low-scoring nature of the game.

The original dataset has been enriched with data about the number of passes received in the attacking third, as it signals the importance of a player in the conclusion of a goal-scoring opportunity, the number of accurate actions under pressure, and the average pass angle under pressure, as these are indicators of the quality of a player while in a dire situation.

Moreover, the sprinting distance of a player is a good predictor of his fatigue. Therefore, it has been added to the augmented dataset, together with the proportion of meters sprinted with respect to the distance between checkpoints and the distance run in the last 15 minutes with respect to the total match distance.

Figure 1 displays the outcome of the PCA applied to the augmented dataset. It is straightforward to understand that the points are not linearly separables. Therefore, the choice of the models to employ should be directed towards non-linear and non-parametric models.

An analysis of the correlation with respect to the target variable shows

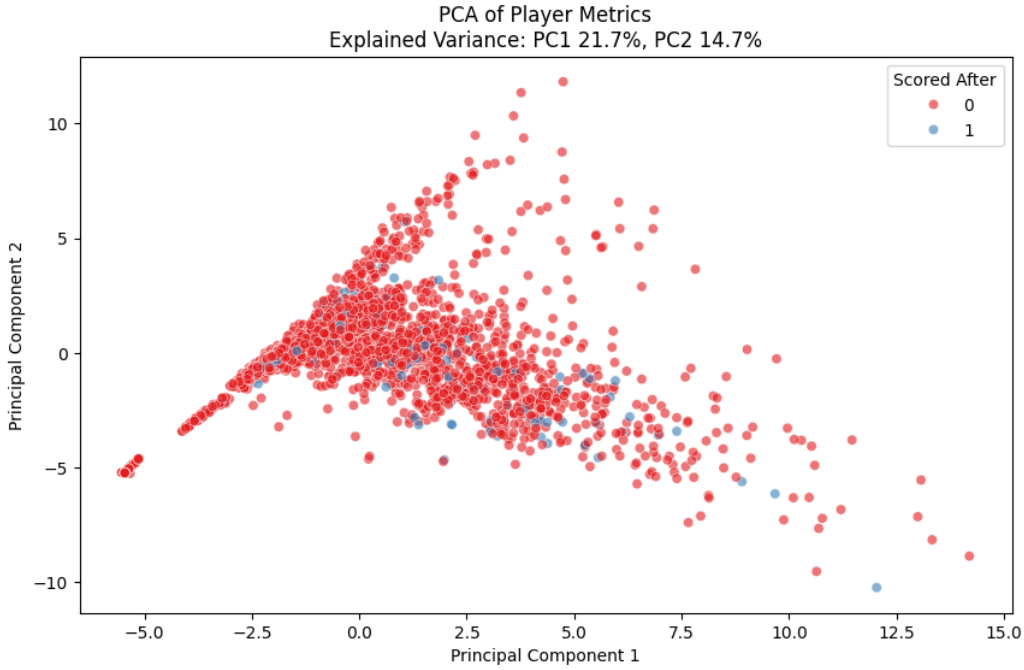


Figure 1: Visualization of the PCA outcome.

the lack of a powerful predictor, as all of the values are rather small, as may be seen in Figure 2.

# 3 Methodology

Three different methodologies have been tried: a stacking ensemble, a classifier based on Gradient Boosting, and a combination of Gradient Boosting and a calibrated classifier.

The stacking ensemble contains a Gradient Boosting classifier with learning rate 0.05 and maximum depth of 3, a two-layered MLP with 13 and 8 hidden nodes, a Gaussian process classifier, and a support vector machine with radial kernel. The main idea behind this choice would be to average out the weaknesses on the submodels to obtained a less biased prediction. In addition to this, as is clear from the exploratory data analysis, the classes are not linearly separable, so heavily non-linear or non-parametric models

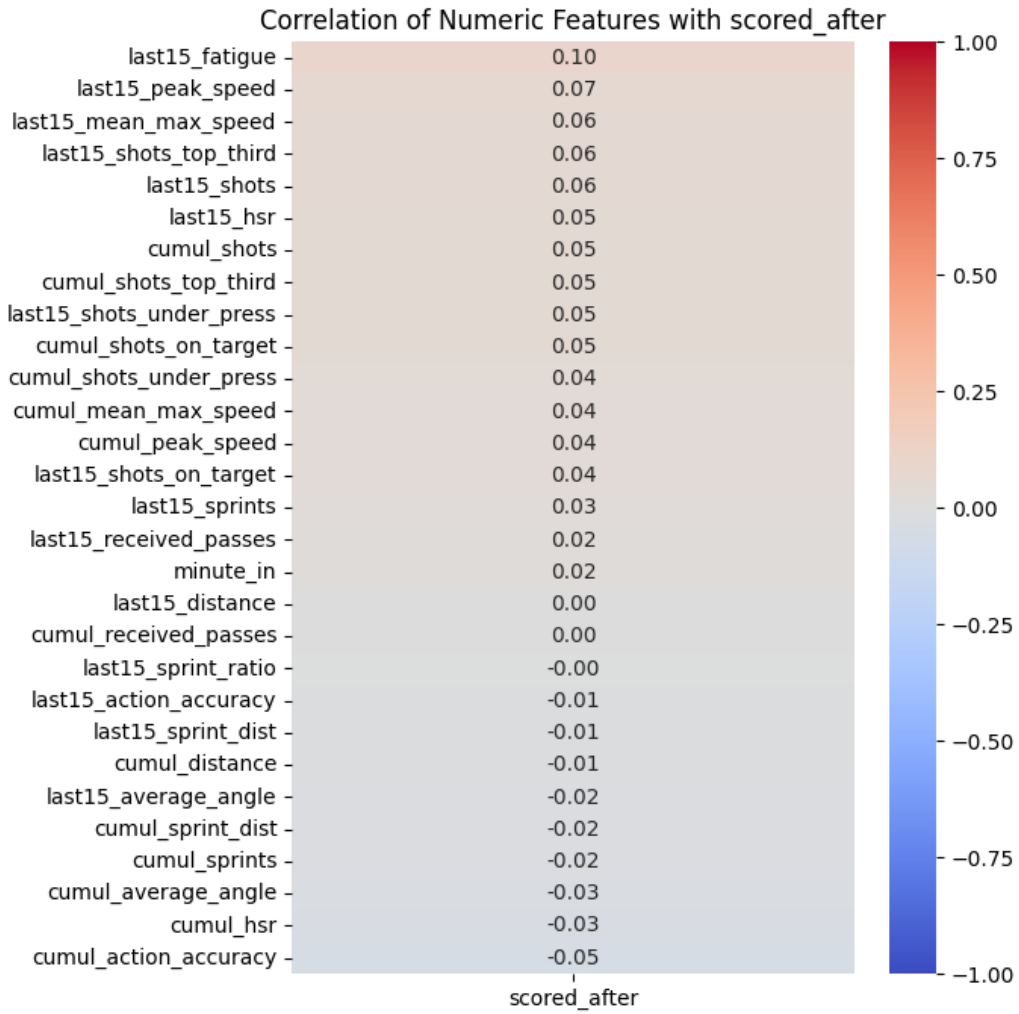


Figure 2: Visualization of the PCA outcome.

should be preferred.

The second approach is based on Gradient Boosting, which is an ensemble learning technique that builds a strong predictive model by sequentially combining multiple weak learners (Friedman, 2001). Unlike techniques that build models independently, Gradient Boosting focuses on correcting the errors of previous iterations. In each step, a new model is trained to predict the residuals of the combined ensemble. By using a gradient descent algorithm to minimize a specific loss function, the model iteratively nudges its predictions in the direction that most reduces error. This additive approach allows the final ensemble to capture complex, non-linear relationships in data with high precision.

The classifier has been endowed with a learning rate of 0.05 and a maximum depth of 3. The optimal class rebalance has been selected via nested cross-validation.

For the third approach, to ensure that the predicted class probabilities accurately reflect empirical likelihoods, the Gradient Boosting classifier was integrated with a post-processing calibration layer. Although Gradient Boosting frameworks like XGBoost demonstrate high discriminative power, their raw output scores are frequently uncalibrated, often resulting in probability estimates that do not align with true class frequencies (Niculescu-Mizil and Caruana, 2005). To mitigate this, isotonic regression, a non-parametric technique that fits a monotonic, piecewise constant function to map the raw classifier scores to calibrated probabilities, was employed (Zadrozny and Elkan, 2002). Unlike parametric alternatives such as Platt scaling, isotonic regression is more flexible as it makes no assumptions regarding the specific functional form of the mapping, making it particularly effective for correcting the complex distortions often found in boosted tree outputs. This calibration ensures that the model's confidence scores are statistically reliable, which is critical for decision-making processes where the magnitude of the probability is as significant as the final classification.

Furthermore, to provide a granular and theoretically grounded interpret-

ation of the model's predictions, we employed SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017). Based on cooperative game theory, SHAP decomposes the output of the machine learning model by assigning each feature a Shapley value, which quantifies its specific contribution to a particular prediction relative to the average baseline. This approach offers a significant advantage over traditional feature importance metrics because it satisfies the properties of consistency and local accuracy, ensuring that the sum of the feature contributions exactly equals the difference between the model's prediction and the expected value. By calculating these additive feature attributions, we were able to transition from a "black-box" model to one that offers both local interpretability and global interpretability, while accurately accounting for interactions between variables.

Only for the stacking ensemble, in order to make the dataset less imbalanced, Synthetic Minority Over-sampling Technique (SMOTE) has been applied (Bowyer et al., 2002). Rather than simply duplicating existing minority class records (which often leads to overfitting), SMOTE creates entirely new, "synthetic" examples by interpolating between existing ones. The algorithm selects a minority instance, identifies its $k$-nearest neighbors, and then generates a new data point at a random location along the line segment connecting the instance to one of its neighbors, via the formula

$$x_{new} = x_i + \lambda \cdot (x_{zi} - x_i), \text{ for } \lambda \in ]0, 1[.$$

The dataset has also been randomly split with the purpose of separating training set and test set, which contains 20% of the observations.

# 4 Results

The models have been evaluated on seven datasets: the base one (B), the augmented one (A), the augmented one without goalkeepers (AG), one with only the rolling covariates (15), one with only the rolling covariates without goalkeepers (15G), one with only the cumulative covariates (C), one with only the cumulative covariates without goalkeepers (CG).

| DF | Stacking | XGB | Calibrated |
|---|---|---|---|
| B | 0.680 | 0.692 | 0.705 |
| A | 0.712 | 0.740 | 0.721 |
| AG | 0.677 | 0.744 | 0.750 |
| 15 | 0.701 | 0.684 | 0.681 |
| 15G | 0.645 | 0.680 | 0.700 |
| C | 0.747 | 0.735 | 0.752 |
| CG | 0.760 | 0.757 | 0.759 |

Table 1: Comparison of the AUCs across models and datasets

It has been decided to use an dominant metric the AUC score, as the F1 score and any form of weighted accuracy is not meaningful for the calibrated model.

Table 1 contains the values of the AUC across different combinations of dataset and model.

The outcome of the SHAP analysis is displayed in Figure 3.

# 5 Discussions and conclusions

Table 1 shows that the most striking trend is the superiority of the C and CG datasets. These datasets consistently yield the highest AUC scores across all models, peaking at 0.760. Conversely, the 15 and 15G datasets represent the lowest performance tier, with AUCs often struggling to surpass the 0.700 threshold. This suggests that the long-term historical context is a much stronger predictor for the target variable than short-term fluctuations.

Removing goalkeepers has a polarized effect depending on the model architecture. For the calibrated model, it consistently improves performance. For instance, in the augmented dataset, the AUC jumps from 0.721 (A) to 0.750 (AG). This likely indicates that goalkeeper-specific data introduces noise or outliers that the isotonic regression calibration struggles to map

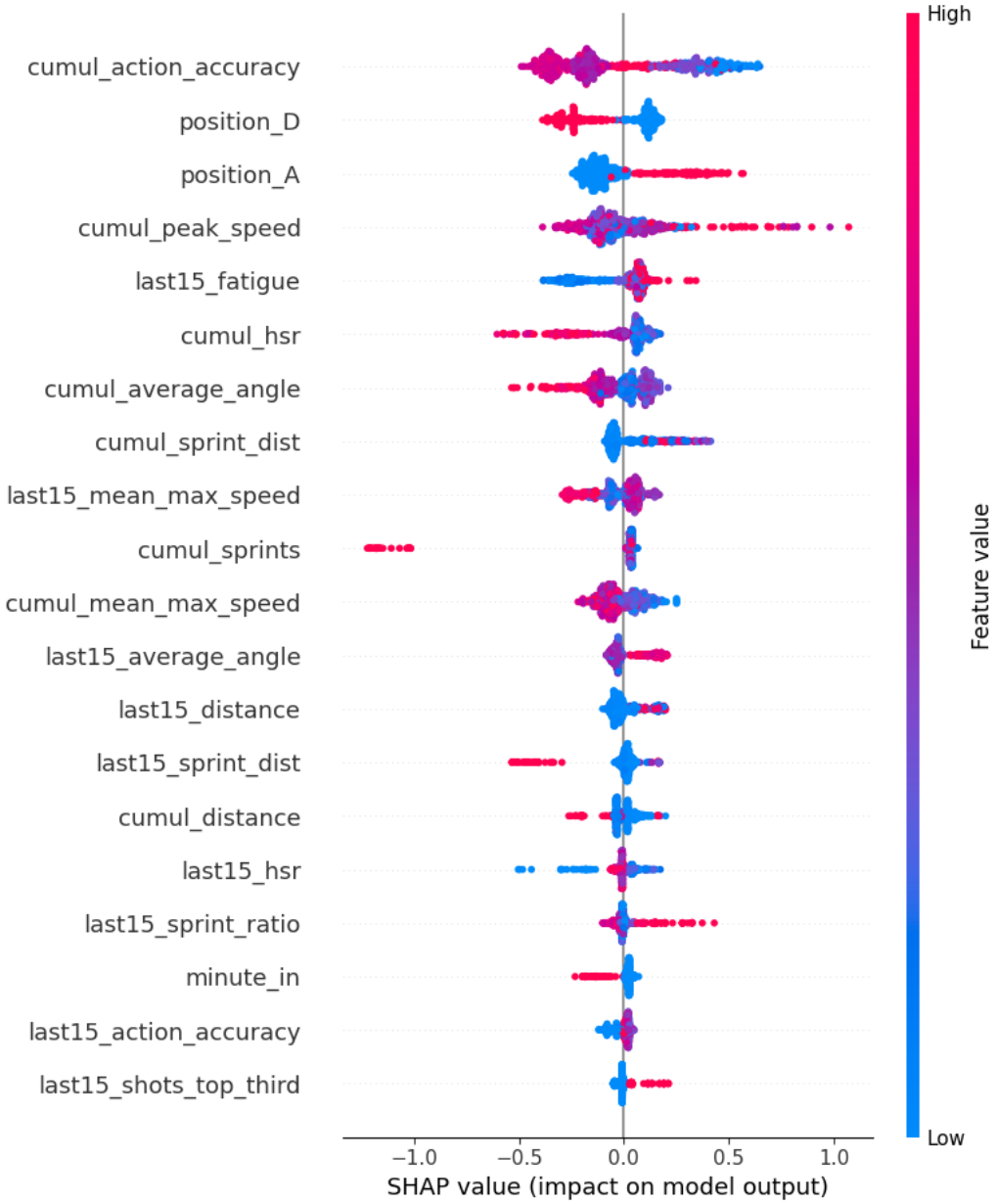


Figure 3: Visualization of the SHAP outcome.

effectively. On the other hand, the stacking ensemble sees significant performance drops when goalkeepers are removed in the augmented (from 0.712 to 0.677) and "Last15" (from 0.701 to 0.645) sets. This suggests the stacking ensemble relies heavily on goalkeeper features to find its predictive edge in those specific feature spaces.

Therefore, the Gradient Boosting classifier is remarkably consistent, maintaining scores in the $0.730 - 0.750$ range for most datasets and avoiding the deep performance dips seen in the stacking ensemble model.

While calibration is usually performed to improve probability reliability rather than discriminative power, it actually outperforms the base Gradient Boosting in 4 out of the 6 augemented datasets (AG, 15G, C, CG). This indicates that the isotonic regression is not just scaling probabilities but is actually refining the ranking of the observations effectively.

The stacking ensemble achieves the highest overall AUC (0.760 on CG) but also the lowest overall AUC (0.645 on 15G). This suggests that, while the ensemble can capture complex relationships in the cumulative data, it is prone to instability when the feature set is restricted or noisy.

Figure 3 clearly shows that some covariates, in particular the positions and the running metrics, are able to give a clear discrimination on the likelihood of scoring a goal. This is due to the fact that fatigue is an important metric on the ability to have an impact on the match. Moreover, also the behaviour under pressure is an intersting predictor, as more quality in the executions appear to have a positive influence on scoring a goal.

In conclusion, assessing the probability of scoring a goal in football is a rather demanding task, but the combination of a non-linear model with high predictive power and the availability of sprinting data is able to provide some useful insights.

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